

Platonic Solids and subgroups of $\mathbf{SO}_3(\mathbb{R})$

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The group of rotational symmetries of a circle S^1 is denoted by $\mathbf{SO}_2(\mathbb{R})$. It is not difficult to see that upon rotating a point $(x, y) \in S^1$ by an angle θ in the anti-clockwise direction, the co-ordinates of the new point are $(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$. Observe that these new co-ordinates can be obtained as the matrix multiplication:

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix}.$$

We denote:

$$R_\theta \stackrel{\text{notn.}}{:=} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Observe that $R_\theta R_\phi = R_{\theta+\phi}$. Thus

$$\mathbf{SO}_2(\mathbb{R}) = \{R_\theta : \theta \in \mathbb{R}\}.$$

This clearly is an Abelian group. Now a question:

Question How many convex regular polygons can be inscribed in a circle?

The answer is trivial, *infinitely many*. The groups of rotations of these polygons are cyclic groups. These are precisely all the subgroups of $\mathbf{SO}_2(\mathbb{R})$. The proof of this is ‘analogous’ to the proof that all additive subgroups of $\mathbb{Z}$ are cyclic.

We try to imitate the above for the *three-dimensional* situation. Let us denote the group of rotational symmetries of a (unit) sphere S^2 by $\mathbf{SO}_3(\mathbb{R})$. This time the situation is not straightforward. There are multiple *axes of rotation*. If we choose the axis as Z -axis and rotate the sphere in the anti-clockwise direction (as seen from the ‘positive’ part the of Z -axis), the corresponding rotation matrix will be:

$$R_\theta^Z = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Similarly for other axes we have:

$$R_{\theta}^X = \begin{pmatrix} 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \end{pmatrix}$$

and

$$R_{\theta}^Y = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}.$$

Observation The group $\text{SO}_3(\mathbb{R})$ is not commutative: $R_{\frac{\pi}{2}}^X R_{\frac{\pi}{2}}^Y \neq R_{\frac{\pi}{2}}^Y R_{\frac{\pi}{2}}^X$.

Question How many regular convex polyhedra can be inscribed inside a sphere?

Well, first one has to explain the meaning of regular convex polyhedron.

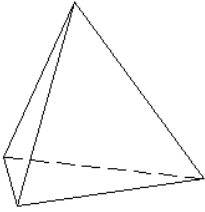
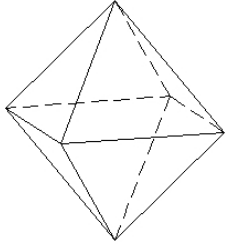
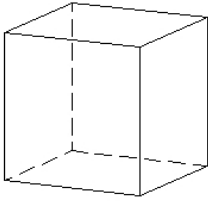
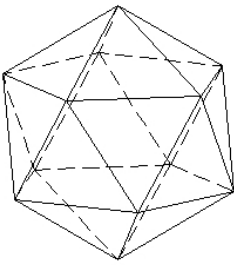
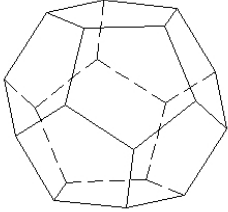
Definition A convex polyhedron is said to be *regular* if:

1. Its faces are convex regular n -gons (for some $n \in \mathbb{N}$, $n \geq 3$) and any two faces are congruent.
2. Only possible intersections of faces are at edges.
3. At each vertex the same number (say m) of faces meet.

It is clear that n and m are at least 3. Regular convex polyhedra are also called *Platonic solids*. Cube is an example of a Platonic solid. For a cube $n = 4$, as each face is a square (regular 4-gon), and $m = 3$ as at each vertex three faces meet.

Definition For a Platonic solid the pair $\{n, m\}$ is called its *Schläfli symbol*. For a tetrahedron the Schläfli symbol is $\{3, 3\}$.

Here is a list of Platonic solids:

Shape	Name	Schläfli symbol
	Tetrahedron	$\{3, 3\}$
	Octahedron	$\{3, 4\}$
	Cube	$\{4, 3\}$
	Icosahedron	$\{3, 5\}$
	Dodecahedron	$\{5, 3\}$

What is amazing is that this is the complete list of Platonic solids. To prove it we shall use the following result from topology/graph theory.

Theorem 1 (Euler) *If a polyhedron has v vertices, f faces and e edges then $v - e + f = 2$.*

Proof The proof is not very difficult. A proof using induction on v and another using dual graphs can be found in the book Basic Topology by M.A. Armstrong. If you have studied Algebraic Topology then you should be able to relate it with the Euler characteristic of the 2-sphere. $\square$

Theorem 2 (Euclid) *There are only five Platonic solids: tetrahedron, cube, octahedron, icosahedron and dodecahedron.*

Proof Let us count the number of edges in two different ways:

1. Since each face consists of n edges and there are f faces, the total number of edges counted twice (as each edge belongs to two faces) is nf . Therefore we conclude that $nf = 2e$.
2. At each vertex m edges meet. Thus the total number of edges counted twice (as each edge corresponds to two vertices) is mv . Thus $mv = 2e$.

We plug in this information in the Euler's formula to get:

$$v - e + f = \frac{2e}{m} - e + \frac{2e}{n} = 2.$$

Therefore

$$\frac{1}{m} + \frac{1}{n} = \frac{1}{2} + \frac{1}{e} > \frac{1}{2}.$$

Since $n \geq 3$ and $m \geq 3$, in view of above inequality, the only possibilities are:

1. $n = m = 3$: In this case $e = 6, v = f = 4$ and *tetrahedron* is the corresponding Platonic solid.
2. $n = 3, m = 4$: In this case $e = 12, v = 6, f = 8$ and *octahedron* is the corresponding Platonic solid.

3. $n = 4, m = 3$: In this case $e = 12, v = 8, f = 6$ and *cube* is the corresponding Platonic solid.
4. $n = 3, m = 5$: In this case $e = 30, v = 12, f = 20$ and *icosahedron* is the corresponding Platonic solid.
5. $n = 5, m = 3$: In this case $e = 30, v = 20, f = 12$ and *dodecahedron* is the corresponding Platonic solid. $\square$

Each of the Platonic solids can be inscribed in a sphere. Any rotational symmetry of these solids gives a rotational symmetry of the sphere and hence the rotational symmetry groups of these solids are subgroups of $\mathbf{SO}_3(\mathbb{R})$.

Exercise Prove that groups of rotational symmetries of Platonic solids are finite. To show this, it is enough to show that there are finitely many axes of rotations and about each axis only finitely many rotations give a symmetry.

We now understand the symmetry groups of Platonic solids.

Tetrahedron

Imagine a tetrahedron is kept on a plane. Enumerate the four vertices as 1, 2, 3 and 4 with the vertex 1 being on the top. There are two types of axes in the tetrahedron:

1. *The ones for which one end of the axis is a vertex*: For these rotations the other end of the axis has to be there at the mid of the opposite face. For the vertex 1 there are two non-trivial rotations. We call them $(2\ 3\ 4)$ and $(2\ 4\ 3)$. Other similar rotations are: $(1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4)$ and $(1\ 4\ 2)$.
2. *The ones for which the axes is the line joining mid points of two edges having no common vertex*: Consider the mid points of the edge $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$. Taking the line joining these two mid points gives an axis of rotation of the tetrahedron. This gives one non-trivial rotation which we denote by $(1\ 2)(3\ 4)$. Other such rotations are: $(1\ 3)(2\ 4)$ and $(1\ 4)(2\ 3)$.

Trivial rotation, the identity, is still missing. Including this too in the list we have the rotation group **Tet** of Tetrahedron as:

$$\mathbf{Tet} = \{1, (2\ 3\ 4), (2\ 4\ 3), (1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4), (1\ 4\ 2), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$$

Incidentally if we write the alternating group A_4 with the cycle notation then the ‘obvious’ map between A_4 and **Tet** turns out to be an isomorphism. A more convincing argument to show the isomorphism $\mathbf{Tet} \simeq A_4$ is the following: Clearly $|\mathbf{Tet}| = 12$. Denote: $s' = (1\ 2)(3\ 4)$ and $t' = (2\ 3\ 4)$. Clearly orders of s' and t' in **Tet** are 2 and 3 three respectively. Now with the help of a tetrahedron try computing the order of $s't'$. What is it? Three. We thus have a homomorphism:

$$\phi : \langle s, t : s^2, t^3, (st)^3 \rangle \rightarrow \mathbf{Tet}.$$

given by $\phi(s) = s', \phi(t) = t'$. This homomorphism is surjective and we know that $|\mathbf{Tet}| = 12$. Further $\phi : \langle s, t : s^2, t^3, (st)^3 \rangle$ is a representation of the group A_4 , which has 12 elements. Thus $\mathbf{Tet} \simeq A_4$.

Cube and Octahedron

The symmetry groups of cube and octahedron are same. This is because they are dual in the sense of ‘reversing’ the roles of faces and vertices. We start with identifying axes for the rotational symmetry of the cube. There are three types of axes:

1. *The ones whose axes pass through the mid points of opposite faces:* There are three such axes. Further for each such axis there are three non-trivial rotations (by 90° , 180° and 270°). Thus the total number of rotations arising this way is: $3 \times 3 = 9$.
2. *The ones whose axes pass through the mid points of opposite edges:* There are six such axes. Further for each such axis there is only one non-trivial rotation (by 180°). Thus the total number of rotations arising this way is: $6 \times 1 = 6$.
3. *The ones whose axes pass through opposite vertices:* There are four such axes. Further for each such axis there are two non-trivial rotations (by 120° and 240°). Thus the total number of rotations arising this way is: $4 \times 2 = 8$.

In the end we put identity, the trivial rotation, to make the total count $9 + 6 + 8 + 1 = 24$.

Let s' denote a non-trivial rotation about an axes of type 2 above and t' denote a rotation of type 3 then the product $s't'$ is a 90° rotation of type 1. This suggests that there is a surjective homomorphism

$$\langle s, t : s^2, t^3, (st)^4 \rangle \rightarrow \mathbf{Cube}.$$

Since $\langle s, t : s^2, t^3, (st)^4 \rangle \simeq S_4$ and **Cube** is of size 24, the above homomorphism is an isomorphism and we have $\mathbf{Cube} \simeq S_4$.

Icosahedron and Dodecahedron

Since icosahedron and dodecahedron are dual, their symmetry groups are same. As before, we start with identifying axes for the rotational symmetry of icosahedron. There are three types of axes:

1. *The ones whose axes pass through the mid points of opposite faces:* There are ten such axes. Further for each such axis there are two non-trivial rotations (by 120° and 240°). Thus the total number of rotations arising this way is: $10 \times 2 = 20$.
2. *The ones whose axes pass through the mid points of opposite edges:* There are fifteen such axes. Further for each such axis there is only one non-trivial rotation (by 180°). Thus the total number of rotations arising this way is: $15 \times 1 = 15$.
3. *The ones whose axes pass through opposite vertices:* There are six such axes. Further for each such axis there are four non-trivial rotations (by 120° and 240°). Thus the total number of rotations arising this way is: $6 \times 4 = 24$.

Along with the identity rotation the total count is $20 + 15 + 24 + 1 = 60$. As in the previous cases, we establish a surjective homomorphism

$$\langle s, t : s^2, t^3, (st)^5 \rangle \rightarrow \mathbf{Ico}$$

and use the isomorphism $\langle s, t : s^2, t^3, (st)^5 \rangle \simeq A_5$ and size count of **Ico** to prove that $\mathbf{Ico} \simeq A_5$.

There is a more direct way to see the isomorphism $\mathbf{Ico} \simeq A_5$.

Ico is simple

First the distribution of 60 elements of icosahedron, according to their orders is:

Order	Number of elements	Axis
1	1	–
2	15	Mid points of edges
3	20	Mid points of faces
5	24	Vertices

Thus we have a partition $60 = 1 + 15 + 20 + 24$. Since any two conjugate elements are have same order, the class equation of **Ico** is a refinement of the above partition. Observe that **Ico** acts transitively on the set of edges of the icosahedron. Thus all the groups of order 2 are conjugate to each other. The same is true for groups of order 3 and 5 - the action of **Ico** on the set of faces and vertices is also transitive. From this it is evident that all 15 elements of order 2 are conjugate to each other.

We claim that all elements of order 3 are also conjugate. Any element of order 3 lies in some subgroup of the form $\text{stab}(F) = \{1, g, g^2\}$ for some face F . If r is the rotation by 180° about an axis orthogonal to the axis passing through the mid point of F then $rgr^{-1} = g^2$. Further, since **Ico** acts transitively on the set of faces of an icosahedron, the groups $\text{stab}(F)$ and $\text{stab}(F')$ are conjugate. From this we conclude that any two elements of order three are conjugate.

Now finally, we see that elements of order 5 split into two conjugacy classes. By the argument as in the case of order 3 elements, rotations by 72° and 288° are conjugate and those by 144° and 216° are conjugate. Also, the rotations by 72° are not conjugate to the ones by 144° , else all 24 elements of order 5 will form a conjugacy classes which is a contradiction in because $24 \nmid 60$.

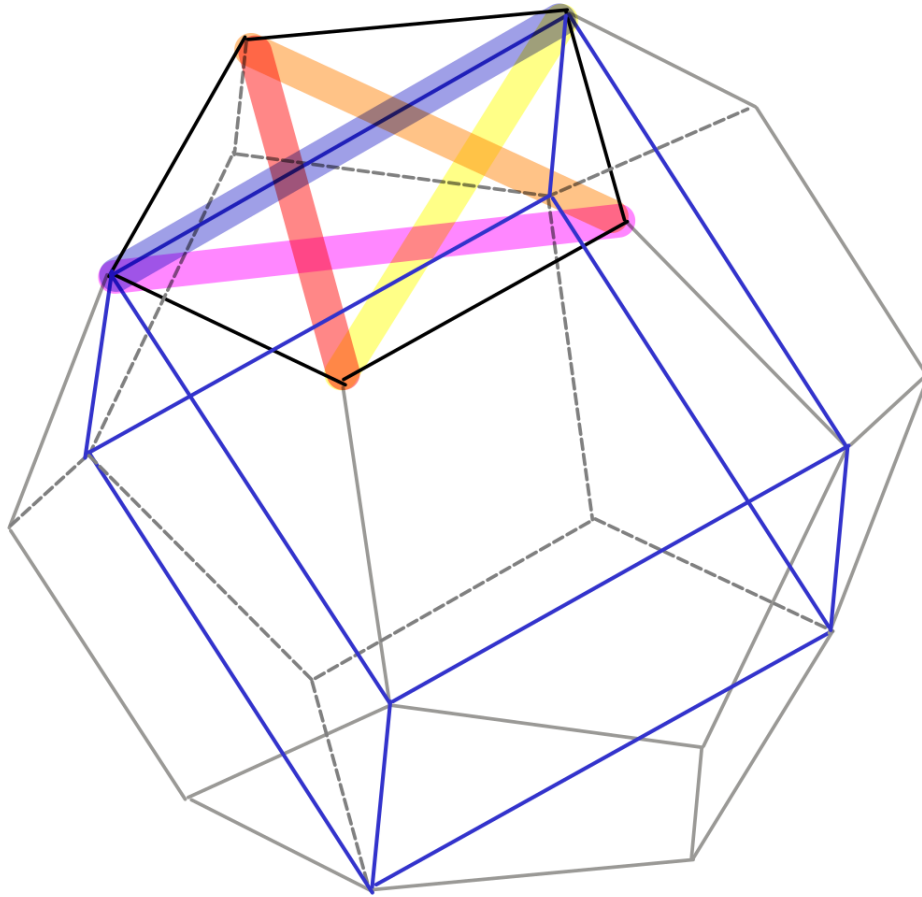
The class equation, therefore, of the group **Ico** is:

$$60 = 1 + 15 + 20 + 12 + 12.$$

If there were normal subgroups in **Ico**, they would be disjoint unions of conjugacy classes along with the identity element. However none of the partial sums (with 1 as a term) in the above class equation divides 60. This shows that the group **Tet** does not have normal subgroups.

$$\mathbf{Ico} \simeq A_5$$

We first show that there is a homomorphism from $\mathbf{Ico} \rightarrow S_5$. For this it is enough to give an action of $\mathbf{Ico}$ on a set of five elements. This set is actually the set of five cubes inscribed in a Dodecahedron. One of the five such cubes is shown in the following figure:



Since $\mathbf{Ico}$ is simple, the homomorphism $\mathbf{Ico} \rightarrow S_5$ is injective. Let G be the image of this homomorphism. We restrict the signature homomorphism $S_5 \rightarrow \{-1, 1\}$ to the subgroup G . If $G \rightarrow \{-1, 1\}$ is surjective then its kernel will be a normal subgroup of order 30, which can not be the case as G is simple, being isomorphic to $\mathbf{Ico}$. Thus $G \subseteq A_5$. Counting the number of elements gives: $\mathbf{Ico} \simeq G = A_5$. $\square$

Finite subgroups of $\mathrm{SO}_3(\mathbb{R})$

In this section we show that Platonic solids and ‘two dimensional’ symmetries provide all finite subgroups of $\mathrm{SO}_3(\mathbb{R})$. The precise statement is:

Theorem 3 Let G be a finite subgroup of $\mathrm{SO}_3(\mathbb{R})$. Then G is isomorphic to one of the following:

C_n : A finite cyclic group.

D_n : A finite Dihedral group.

V_4 : Klein’s 4-group.

A_4 : Group of rotational symmetries of a tetrahedron.

S_4 : Group of rotational symmetries of a cube/octahedron.

A_5 : Group of rotational symmetries of an icosahedron/dodecahedron.

(Here C_n appears because of $\mathrm{SO}_2(\mathbb{R})$ and D_n appears because it lies in $\mathrm{O}(2)$ and ‘the two dimensional’ reflections can be treated as rotations in three dimensions. Last four groups ‘genuinely’ come from three dimensional rotations.)

Proof Let G be a finite subgroup of $\mathrm{SO}_3(\mathbb{R})$. Since each $g \in G$ is a rotation, we denote by ℓ_g the corresponding axis of rotation. Consider the set:

$$X = \{\ell_g \cap S^2 : g \in G\},$$

where S^2 denotes the unit sphere. The set X is just the collection of the two ends of the axes about which some rotation lies in G . Elements of X will be called *poles*. Both the elements in the intersection $\ell_g \cap S^2$ are called *poles of g* . Since G is finite, there are finitely many poles. If G is a cyclic group then X consists of just two antipodal points. In what follows, we shall identify all the possibilities of the size $|X|$ and the distribution of poles on the sphere.

First an observation: *the group G acts on X* . To justify this, let $P \in X$ be a pole of ℓ_h for some $h \in G$. Then $hP = P$. Further for an arbitrary $g \in G$ we have:

$$ghg^{-1}(gP) = ghP = g(hP) = gP.$$

Thus the point $gP \in S^2$ is fixed by the element $ghg^{-1} \in G$, and hence gP is a pole of $\ell_{ghg^{-1}}$. We therefore conclude that G acts on X .

We now investigate into the distribution of poles on the sphere and their orbits under G -action. Let m denotes the number of orbits of this action then from Burnside's formula we have:

$$n = \frac{1}{|G|} (|X| + (|G| - 1)2). \quad (1)$$

On the right side of this equation, the additive term $|X|$ signifies that the identity $1 \in G$ keeps all poles fixed and $(|G| - 1)2$ signifies that each non-identity element $g \in G$ fixes only two elements, which are $\ell_g \cap S^2$. For each orbit of the G -action on X we choose representatives, say $x_1, x_2, \dots, x_n$. We then rewrite the above summation as:

$$n = \frac{1}{|G|} \left(\sum_{i=1}^n |\text{orbit}(x_i)| + (|G| - 1)2 \right).$$

Rearranging we get:

$$\begin{aligned} 2 \left(1 - \frac{1}{|G|} \right) &= n - \frac{1}{|G|} \sum_{i=1}^n |\text{orbit}(x_i)| \\ &= n - \sum_{i=1}^n \frac{|\text{orbit}(x_i)|}{|G|} \\ &= n - \sum_{i=1}^n \frac{1}{|\text{stab}(x_i)|} \\ &= \sum_{i=1}^n \left(1 - \frac{1}{|\text{stab}(x_i)|} \right) \end{aligned} \quad (2)$$

Assuming the group G is non-trivial, i.e. $|G| \geq 2$, in view of the above equation we have:

$$1 \leq 2 \left(1 - \frac{1}{|G|} \right) < 2.$$

Further observe that for each x_i we have $|\text{stab}(x_i)| \geq 2$ because $\text{stab}(x_i)$ contains $1 \in G$ and at least a rotation with x_i as a pole. From this we conclude that:

$$\frac{1}{2} \leq \left(1 - \frac{1}{|\text{stab}(x_i)|} \right) < 1.$$

Taking summation over all orbits:

$$\frac{n}{2} \leq \sum_{i=1}^n \left(1 - \frac{1}{|\text{stab}(x_i)|}\right) < n.$$

Therefore

$$\frac{n}{2} \leq 2 \left(1 - \frac{1}{|G|}\right) < n.$$

The first inequality gives:

$$\frac{n}{2} \leq 2 \left(1 - \frac{1}{|G|}\right) < 2.$$

Therefore $n \leq 4$. The second inequality and the fact that $|G| \geq 2$ give:

$$\frac{1}{2} \leq 1 - \frac{1}{|G|} < \frac{n}{2}.$$

Thus $n > 1$ and we conclude that there are only two possibilities: $n = 2$ and $n = 3$. We now do case by case analysis:

Case $n = 2$: Using the formula 1 we get $|X| = 2$. Thus there are only two poles x_1 and x_2 , both in different orbits. This can happen only when G consists of rotations with the line $x_1 \leftrightarrow x_2$. Therefore the group G is cyclic.

Case $n = 3$: There are three orbits of the action of G on poles. We rename the three orbit representatives as x, y and z . Using formula 2 we get:

$$1 < 1 + \frac{2}{|G|} = \frac{1}{|\text{stab}(x)|} + \frac{1}{|\text{stab}(y)|} + \frac{1}{|\text{stab}(z)|} \quad (3)$$

If all three stabilizers are of order 3 or more then the sum $\frac{1}{|\text{stab}(x)|} + \frac{1}{|\text{stab}(y)|} + \frac{1}{|\text{stab}(z)|}$ is at most 1, which is a contradiction to the formula 3. Thus one of the stabilizers, say $\text{stab}(x)$, has order 2. It is now an easy exercise to confirm that the only possibilities are:

- i. $|\text{stab}(x)| = 2, |\text{stab}(y)| = 2, |\text{stab}(z)| = r$ for some $r \geq 2$.
- ii. $|\text{stab}(x)| = 2, |\text{stab}(y)| = 3, |\text{stab}(z)| = 3$.
- iii. $|\text{stab}(x)| = 2, |\text{stab}(y)| = 3, |\text{stab}(z)| = 4$.
- iv. $|\text{stab}(x)| = 2, |\text{stab}(y)| = 5, |\text{stab}(z)| = 5$.

Subcase i: $|\text{stab}(x)| = 2$, $|\text{stab}(y)| = 2$, $|\text{stab}(z)| = r$ for some $r \geq 2$.

If $r = 2$ then plugging in the values of orders of stabilizers in the formula 3 we get $|G| = 4$. There are only two group of order 4, Cyclic group C_4 and the Klein's 4-group V_4 . Each of them occurs as a subgroup of $\text{SO}_3(\mathbb{R})$. In fact C_4 is a subgroup of $\text{SO}_2(\mathbb{R})$ as we have already seen. To see that V_4 is a subgroup of $\text{SO}_3(\mathbb{R})$, imagine rotations by 180° about 'standard' X , Y and Z axes. These four rotations along with identity form a subgroup of $\text{SO}_3(\mathbb{R})$ isomorphic to V_4 .

If $r \geq 3$ then plugging in the values of orders of stabilizers in the formula 3 we get $|G| = 2r$. Since $\text{stab}(z)$ fixes the pole z (and hence the antipodal pole $-z$), the group $\text{stab}(z)$ is a cyclic group (of order r). We write:

$$\text{stab}(z) = \{1, g, g^2, \dots, g^{r-1}\}.$$

We now observe how this cyclic group acts on other poles - take the pole x as a sample case. It is not difficult to see that all poles $x, g(x), g^2(x), \dots, g^{n-1}(x)$ are all distinct, otherwise $g^{i-j}(x) = x$ for some $i \neq j$ will suggest that g^{i-j} fixes x , but elements of $\text{stab}(z)$ fix z and $-z$ only. Moreover $x \neq -z$ as their stabilizers have different orders.

Since $g \in G$ is distance preserving, it is immediate that:

$$|x - g(x)| = |g(x) - g^2(x)| = \dots = |g^{n-1}(x) - x|.$$

and

$$|z - x| = |z - g(x)| = |z - g^2(x)| = \dots = |z - g^{n-1}(x)|.$$

This gives a big clue about the configuration of poles on the sphere: $g^i(x)$ are coplanar, their plane is orthogonal to the axis $z \leftrightarrow -z$ and they form the vertices of a regular r -gon. Call this polygon P .

Observe that the orbit of x is precisely the set $\{x, g(x), g^2(x), \dots, g^{n-1}(x)\}$. This can be seen by the orbit-stabilizer formula and the fact that $|\text{stab}(x)| = 2$. Therefore G maps P to P and we have a homomorphism:

$$\psi : G \rightarrow \text{rot}_3(P),$$

where $\text{rot}_3(P)$ denotes the three dimensional rotational symmetry group of P . What is the kernel of ψ ? Since every non-trivial element $s \in G$ fixes only two points on the sphere, it can't fix whole of P . Thus for a non-trivial $s \in G$ the image $\psi(s)$ is non-trivial and $\ker(\psi)$ is trivial. Therefore $\psi : G \rightarrow \text{rot}_3(P)$ is injective. Since reflections in two dimensions

correspond to certain rotations in three dimensions, it is easy to see that $\text{rot}_3(P) \simeq D_r$ and we have an injective homomorphism $G \rightarrow D_r$. Since $|G| = 2r = |D_r|$, we conclude that $G \simeq D_r$.

Subcase ii: $|\text{stab}(x)| = 2$, $|\text{stab}(y)| = 3$, $|\text{stab}(z)| = 3$.

Using formula 3 we calculate that $|G| = 12$. Orbit-stabilizer formula then gives $|\text{orbit}(z)| = \frac{12}{|\text{stab}(z)|} = 4$. We choose an element $w \in \text{orbit}(z)$, such that $w \notin \{z, -z\}$. Such a choice is possible because $|\text{orbit}(z)|$ is 'big enough'.

Let $\text{stab}(z) = \{1, g, g^2\}$. Then the poles $w, g(w)$ and $g^2(w)$ are all distinct, otherwise $\text{stab}(z)$ will stabilize w which is not possible as $w \notin \{z, -z\}$. We therefore conclude that $\text{orbit}(z) = \{z, w, g(w), g^2(w)\}$. As in the subcase i; the poles $w, g(w)$ and $g^2(w)$ are equidistant from z and form the vertices of an equilateral triangle. Since an orbit is always preserved under the action of G , we choose some $h \in \text{stab}(w)$ and conclude that h permutes other three elements $z, g(w)$ and $g^2(w)$ of the orbit. Further, as before, $z, g(w)$ and $g^2(w)$ are equidistant from w . Thus $\{z, w, g(w), g^2(w)\}$ form a regular tetrahedron and we have a homomorphism:

$$\phi : G \rightarrow \mathbf{Tet}.$$

Since no non-trivial rotation fixes the whole tetrahedron, $\ker(\psi)$ is trivial and a count of elements of G and $\mathbf{Tet}$ gives $G \simeq \mathbf{Tet}$. Recall that $\mathbf{Tet} \simeq A_4$. Thus $G \simeq A_4$.

Subcase iii: $|\text{stab}(x)| = 2$, $|\text{stab}(y)| = 3$, $|\text{stab}(z)| = 4$

Using formula 3 we calculate $|G| = 24$ and from the orbit-stabilizer formula we have $|\text{orbit}(z)| = 6$. We choose $w \in \text{orbit}(z)$ with $w \notin \{z, -z\}$. Let $\text{stab}(z) = \{1, g, g^2, g^3\}$. As in the previous subcases, $w, g(w), g^2(w)$ and $g^3(w)$ are equidistant from z and form the vertices of a square. We observe that $-z \notin \text{orbit}(x) \cup \text{orbit}(y)$. This is because the $|\text{stab}(-z)| = |\text{stab}(z)|$ is different from both $|\text{stab}(x)|$ and $|\text{stab}(y)|$. Thus $-z \in \text{orbit}(z)$ and we have:

$$\text{orbit}(z) = \{z, -z, w, g(w), g^2(w), g^3(w)\}.$$

Again by the same argument since $|\text{stab}(-w)| = |\text{stab}(w)|$ is different from both $|\text{stab}(x)|$ and $|\text{stab}(y)|$, we have $-w \in \text{orbit}(z)$. By choice $-w \notin \{z, -z, w\}$. Thus $-w \in \{g(w), g^2(w), g^3(w)\}$. Since $|w - g(w)| = |g(w) - g^2(w)| = |g^2(w) - g^3(w)| = |g^3(w) - w|$ and $w, g(w), g^2(w), g^3(w)$ form vertices of a square, we have that $g^2(w)$ is antipodal to w . Thus $-w = g^2(w)$ and $-g(w) = g^3(w)$. Thus six points $\{z, -z, w, g(w), g^2(w), g^3(w)\}$

form the vertices of a regular octahedron. As in the previous subcases, we have an injective homomorphism $G \rightarrow \mathbf{Cube} \simeq S_4$ which is surjective in view of the element count in groups G and S_4 .

Subcase iv: $|\text{stab}(x)| = 2$, $|\text{stab}(y)| = 3$, $|\text{stab}(z)| = 5$

By formula 3 we have $|G| = 60$ and from the orbit-stabilizer formula we get $|\text{orbit}(z)| = 12$. Thus $\text{stab}(z) = \{1, g, g^2, g^3, g^4\}$. Choose $w \in \text{orbit}(z)$ such that $w \notin \{z, -z\}$. Then $\{z, w, g(w), g^2(w), g^3(w), g^4(w)\} \subseteq \text{orbit}(z)$. Moreover counting the orders of various stabilizers, as in the earlier case, we have $-z \in \text{orbit}(z)$. Since $\text{orbit}(z)$ is big enough we can choose $v \in \text{orbit}(z)$ such that $v \notin \{z, -z, w, g(w), g^2(w), g^3(w), g^4(w)\}$. Thus

$$\{z, -z, w, g(w), g^2(w), g^3(w), g^4(w), v, g(v), g^2(v), g^3(v), g^4(v)\}.$$

These poles form vertices of a regular tetrahedron. Now argue as in earlier subcases to show that $G \simeq \mathbf{Ico} \simeq A_5$.

Some Classics

The first account of finite subgroups of $\mathbf{SO}_3(\mathbb{R})$ is found in Felix Klein's 1884 classic work *Lectures on the icosahedron*. This book is now available with Dover Publication. *Introduction to Geometry* by H.S.M. Coxeter is a place to learn more about Platonic solids.